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EMPLOYMENT

Auburn University Fall 2024-
Postdoctoral Fellow

Simons Laufer Mathematical Sciences Institute Spring 2024
Postdoctoral Fellow

University of Arkansas 2020-2023
Visiting Assistant Professor

EDUCATION

Ph.D. Mathematics 2015-2020
University of Kansas

Advisor: Hailong Dao

Thesis: Homological Properties of Structures in Commutative Algebra and Algebraic Combinatorics

M.A. Mathematics 2013-2015
Wake Forest University

Advisor: Frank Moore

Thesis: Noncommutative Complete Isolated Singularities

B.S. Mathematics (magna cum laude) 2009-2013
Berry College

PUBLISHED AND ACCEPTED WORKS

- [1] E. Celikbas, O. Celikbas, J. Lyle, R. Takahashi, Y. Yao, On a generalization of Ulrich modules and its applications, *Tohoku Mathematical Journal* (2024), <https://arxiv.org/abs/2308.06606>.
- [2] J. Lyle, J. Montaña, K. Sather-Wagstaff, Exterior powers and Tor-persistence, *Journal of Pure and Applied Algebra* (2022), <https://www.sciencedirect.com/science/article/pii/S0022404921002310>
- [3] B. Holmes, J. Lyle, Rank selection and depth conditions for balanced simplicial complexes, *Electronic Journal of Combinatorics* (2021), <https://www.combinatorics.org/ojs/index.php/eljc/article/view/v28i2p28>.
- [4] J. Lyle, J. Montaña, Extremal growth of Betti numbers and trivial vanishing of (co)homology, *Transactions of the American Mathematical Society*, Volume 373 (2020), pp. 7937-7958. <https://www.ams.org/journals/tran/0000-000-00/S0002-9947-2020-08189-6/>
- [5] T. Kobayashi, J. Lyle, R. Takahashi, On the number of indecomposable Cohen–Macaulay modules which are not locally free on the punctured spectrum, *Journal of Pure and Applied Algebra*, Volume 224 (2020) no. 7. <https://www.sciencedirect.com/science/article/pii/S0022404920300086>

- [6] H. Dao, J. Doolittle, J. Lyle, Minimal Cohen-Macaulay simplicial complexes, *SIAM J. Discrete Math.* (2020), 34(3), 1602-1608. (7 pages) <https://epubs.siam.org/doi/abs/10.1137/19M1275164>
- [7] H. Dao, J. Doolittle, K. Duna, B. Goeckner, B. Holmes, J. Lyle, Higher nerves of simplicial complexes, *Algebraic Combinatorics*, Volume 2 (2019) no. 5, pp. 803-813. https://alco.centre-mersenne.org/item/ALCO_2019__2_5_803_0
- [8] H. Dao, M. Eghbali, J. Lyle, Hom and Ext, revisited, *Journal of Algebra*, Volume 571 (2019), pp. 75-93. <https://doi.org/10.1016/j.jalgebra.2018.12.006>.

SUBMITTED WORKS

- [1] J. Lyle, P. Mantero, Ordinary and Symbolic Powers of Matroids via Polarization, preprint (2024).
- [2] J. Lyle, S. Maitra, Annihilators of (co)homology and their influence on the trace ideal, preprint (2024), <https://arxiv.org/abs/2409.04686>.
- [3] J. Lyle, Generalized trace Submodules and centers of endomorphism rings, preprint (2023), <https://arxiv.org/abs/2311.00220>.
- [4] K. Kimura, J. Lyle, Y. Otake, R. Takahashi, On the vanishing of Ext modules over a local unique factorization domain with an isolated singularity, preprint (2023), <https://arxiv.org/abs/2310.16599>.
- [5] J. Lyle, Endomorphism algebras over commutative rings and torsion in self tensor products, preprint (2023), <https://arxiv.org/abs/2310.14134>.

TEACHING

Auburn University

- Math 1680 (Calculus with Business), Fall 2024

University of Arkansas

- Math 3083 (Linear Algebra), Spring 2023
- Math 2043 (Survey of Calculus), Fall 2021, Spring 2022, Fall 2022
- Math 2564 (Calculus II), Fall 2020, Spring 2021

University of Kansas

- Math 126 (Calculus II), Recitation Instructor, Spring 2019
- Math 104 (Precalculus), Instructor, Fall 2016, Fall 2018
- Math 115 (Applied Calculus), Instructor, Fall 2017, Spring 2018
- Math 127 (Calculus III), Recitation Instructor, Spring 2017

Wake Forest

- Math 121 (Linear Algebra), Teaching Assistant, Spring 2015
- Math 117 (Discrete Mathematics), Teaching Assistant, Fall 2014
- Math 113 (Multivariable Calculus), Teaching Assistant, Spring 2014
- Math 111 (Calculus I), Teaching Assistant, Fall 2013

AWARDS

Paul F. Conrad Scholarship 2019	University of Kansas
Awarded annually to one or two mathematics graduate students for academic merit	
Chancellor's Fellowship 2015-2020	University of Kansas
Department Summer Research Scholarship 2015-2019	University of Kansas
Outstanding Graduate Student Award 2014	Wake Forest
Barton Mathematics Award 2013	Berry College

INVITED TALKS

AMS Spring Southeastern Sectional Meeting <i>San Francisco, CA</i>	May 2024
“Counterexamples for Several Open Problems on the Vanishing of Ext and Tor”	
AMS Spring Southeastern Sectional Meeting <i>San Francisco, CA</i>	May 2024
“Generalized Trace Submodules and Endomorphism Rings”	
SLMath Commutative Algebra Seminar <i>Berkeley, CA</i>	March 2024
“Counterexamples for Several Open Problems on the Vanishing of Ext and Tor”	
Joint Math Meetings <i>San Francisco, CA</i>	January 2024
“Endomorphism Algebras Over Commutative Rings and Torsion in Tensor Products”	
University of Utah Commutative Algebra Seminar <i>Salt Lake City, UT</i>	November 2023
“Counterexamples for Several Open Problems on the Vanishing of Ext and Tor”	
AMS Spring Southeastern Sectional Meeting <i>Cincinnati, OH</i>	March 2023
“Symbolic Powers of Squarefree Monomial Ideals, Revisited”	
AMS Spring Southeastern Sectional Meeting <i>Atlanta, GA</i>	March 2023
“Stable Theory for Complexes and the Auslander-Reiten Conjecture”	
West Virginia University Algebra Seminar <i>Virtual Session</i>	November 2022
“Stable Theory for Complexes and the Auslander-Reiten Conjecture”	
Auburn University Algebra Seminar <i>Auburn, AL</i>	November 2022
“Stable Theory for Complexes and the Auslander-Reiten Conjecture”	
Oklahoma State University Combinatorics and Commutative Algebra Seminar <i>Virtual Session</i>	April 2022
“Symbolic Powers of Squarefree Monomial Ideals, Revisited”	
University of Michigan Commutative Algebra Seminar <i>Virtual Session</i>	April 2022
“Generalizations of Ulrich Modules and Rigidity Theorems”	
Clemson University Algebra and Discrete Math Seminar <i>Virtual Session</i>	November 2021
“Recent progress on the Auslander-Reiten and Huneke-Weigand Conjectures”	
Iowa State University Commutative Algebra Seminar <i>Virtual Session</i>	April 2021
“Constructions in Combinatorial Commutative Algebra, Old and New”	

AMS Special Session on Commutative Algebra <i>Virtual Session</i> “Rank Selection and Depth Conditions for Squarefree Monomial Ideals”	April 2021
Wake Forest University Colloquium <i>Virtual Session</i> “Constructions in Combinatorial Commutative Algebra, Old and New”	March 2021
AMS Special Session on Homological Methods in Algebra <i>Virtual Session</i> “Tor-persistence and related conjectures”	October 2020
AMS Special Session on Commutative Algebra <i>Virtual Session</i> “Maximal Cohen-Macaulay modules that are not locally free on the punctured spectrum”	April 2020
University of Nebraska, Lincoln Commutative Algebra Seminar <i>Lincoln, NE</i> “Extremal growth of Betti numbers and rigidity of (co)homology”	March 2020
University of Nebraska, Lincoln Commutative Algebra Seminar <i>Lincoln, NE</i> “Maximal Cohen-Macaulay modules that are not locally free on the punctured spectrum”	March 2020
Purdue Commutative Algebra Seminar <i>West Lafayette, IN</i> “Generalizations of Ulrich Modules and Rigidity Theorems”	October 2019
AMS Special Session on Homological and Characteristic $p > 0$ Methods in Commutative Algebra <i>Madison, WI</i> “Generalizations of Ulrich Modules and Rigidity Theorems”	September 2019
KUMUNU Jr. <i>University of Nebraska, Lincoln NE</i> “Cohen-Macaulay Rings of Finite CM_+ -Representation Type”	March 2019
AMS Special Session on Commutative and Combinatorial Algebra <i>Auburn University, Auburn AL</i> “Cohen-Macaulay Rings of Finite CM_+ -Representation Type”	March 2019
AMS Special Session on Commutative Algebra <i>University of Arkansas, Fayetteville AR</i> “ c -Ulrich Modules”	November 2018
AMS Special Session on Multiplicities and Volumes: An Interplay Among Algebra, Combinatorics, and Geometry <i>Ann Arbor, MI</i> “ c -Ulrich Modules”	October 2018
KUMUNU Jr. <i>University of Nebraska, Lincoln NE</i> “Hom and Ext, Revisited”	April 2018

SERVICE

- Co-organizer of the Career Development Series at SLMath, 2024
- Co-organizer of the Commutative Algebra Seminar at the University of Arkansas, 2020-2023.
- Co-creator and Co-organizer of CARES, 2020-2022: CARES is a regional seminar series uniting Arkansas, Kansas, Missouri, Nebraska, and Oklahoma State, and which focuses on expository talks given by graduate students. I served, along with Alessandra Costantini and Kyle Maddox, as a global advisor and organizer to the local organizers, who are senior graduate students at their respective institutions.
- Referee for:
 - Rocky Mountain J. Math

- European Journal of Combinatorics
- Communications in Algebra
- AMS Graduate Series in Math
- Journal of the Mathematical Society of Japan
- Bulletin of the London Mathematical Society
- Proceedings of the AMS
- Journal of Algebra and its Applications
- Journal of Commutative Algebra
- Journal of Algebra